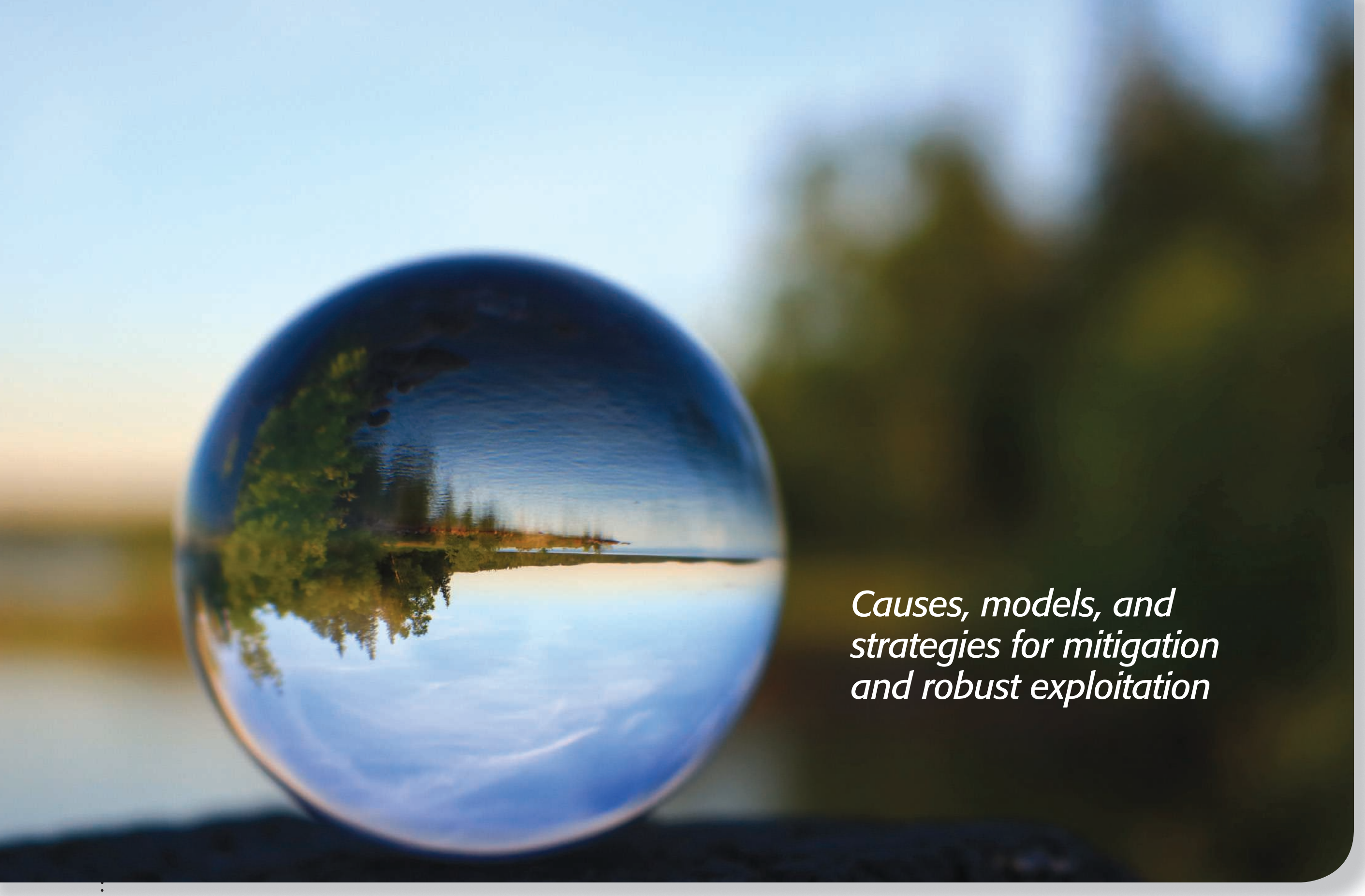

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Spectral Variability of Remotely Sensed Target Materials

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The central problem in hyperspectral remote sensing is characterizing the material components of a scene based on the spectral radiance observed in the image pixels. What makes this challenging is the fact that the spectral response for a given material exhibits considerable variability from a variety of causes: intrinsic (depending on the composition or morphology of the material), extrinsic (depending on the size of an object or the concentration of the material), or environmental (due to illumination, atmospheric distortion, and so on). In this article, we survey many of the causes of spectral variability, describe spectral models for this variability, and outline some signal processing and target detection strategies for analyzing hyperspectral data in a way that is more robust to this variability.

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HYPERSPECTRAL IMAGING

Many remote sensing applications require locating a specific target object or material within a scene, which may then be further identified, characterized, or quantified. A geologist might care about a particular mineral, a forester might want to identify and monitor certain vegetation species, a public-health agent might want to know about particular pollutants and dangerous toxins in the environment, a climate scientist might want to locate sources of methane or carbon dioxide, or a government might want information about nuclear proliferation activities in another country.

With its exquisite discrimination ability, enabled by upward of hundreds of distinct spectral channels in every pixel, hyperspectral imaging has the potential to solve all of these problems. Underlying the many algorithms developed for exploiting hyperspectral imagery is the notion that each material is unequivocally characterized by its unique spectral signature [embodied by the spectral reflectance in the visible

(VIS), near infrared (NIR), and short-wave infrared (SWIR) or by the spectral emittance in the long-wave infrared (LWIR)].

Nonetheless, the notion of a single predetermined spectral signature for each material is an ideal concept not observed in real-world applications. Even in precisely controlled laboratory experiments, the measured spectral reflectance of the same material exhibits some variability. Matters get even worse in hyperspectral remote sensing where target materials are imaged by a sensor mounted on a platform, because many environmental and geometric aspects are involved, and they unavoidably affect the measured spectral radiance for the given material. All of these sources of variability will alter the spectral intensity and shape of a measured spectrum and will affect the performance of hyperspectral image exploitation, particularly in the case of target detection.

The purpose of this survey is to examine the major causes of spectral variability in target materials, survey some associated physical models, and discuss how hyperspectral analyses can be made more robust to this variability. Although there are many kinds of hyperspectral-image exploitation algorithms (e.g., classification, clustering, and unmixing) [1] and virtually all of them are affected by spectral variability, we will concentrate on the problem of target detection and, more specifically, the detection, identification, and location of particular materials of interest.

SOURCES OF TARGET SPECTRAL VARIABILITY

I don't like sand. It's coarse and rough and irritating and it gets everywhere.

—Anakin Skywalker

The most obvious and immediate cause of variability in a target spectrum is the intrinsic variability in the target material itself. Unless the material is a pure substance, there is likely some variability in its composition; for instance, we can refer to "sand" as a single material, but there are many kinds of sand, and any given kind of sand is composed of varying ratios of mineral components [2]. "Pure" materials can undergo chemical variation (e.g., due to oxidation or hydration), and even a chemically pure material with fixed optical properties can exhibit spectral reflectances that vary with material morphology. In addition to these intrinsic variability sources, the effective reflectance at a pixel can depend on extrinsic properties of the material, such as concentration, thickness, or size relative to the pixel. Environmental effects, although not affecting the target reflectance itself, can have a large impact on observed at-sensor spectral radiance; these effects include atmosphere, illumination, adjacent materials, and issues within the sensor itself.

The ultimate consequence of all these sources of variability is that the observed target spectrum in the field often differs from spectral reflectance or emissivity measured in the laboratory and archived in spectral databases, such as the U.S. Geological Survey Spectral Library [3]. The first step in bridging this gap is to identify and catalog these individual sources, while at the same time recognizing that their effects occur together and superimpose on each other. If the sources are understood, then models can be formulated, and from these models algorithms can be designed to achieve some measure of robustness to this variability.

INTRINSIC TARGET VARIABILITY

The most direct source of spectral variability in a target material is due to the chemistry or morphology inherent to the material. When in powdered form, "pure" solid materials become particularly complicated because their spectra are highly variable [4] depending on particle size (e.g., see Figure 1), particle shape, and packing density; this can be further complicated in the case of fine-grained (or "intimate") mixtures of multiple materials. Under ideal conditions, however, the fundamental reflectance properties of a pure solid material are directly characterized by its optical constants (refraction index n and extinction coefficient k) [5], [6]; from those values, one can derive modeled reflectance signatures based on morphology (e.g., for particle size [7], [8] or packing density [9], [10]). As sensor technology has improved in recent years, there has been an increasing need for such reflectance models in order to generate simulated spectra, because obtaining comprehensive spectral measurements under the multitude of possible morphologies is intractable [11], [12]. For solid materials, both pure and mixed, this reflectance also depends on the angle from which the material is illuminated as well as the angle from which it is observed. Those dependencies are encapsulated in the bidirectional reflectance distribution function (BRDF) [13]–[15], which is a function of four variables that define how light reflects off of an opaque surface [16]:

$$r_{\mathrm{BRDF}} = \frac{L(\theta_O, \phi_O)}{E(\theta_i, \phi_i)}[sr^{-1}]. \tag{1}$$

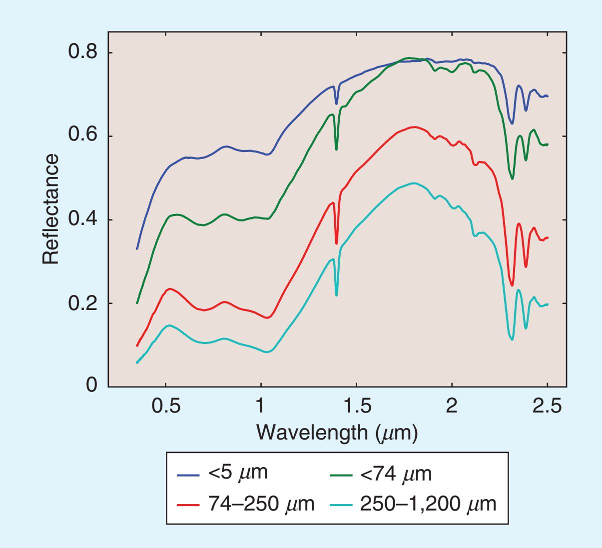

FIGURE 1. Four spectra of the mineral actinolite, based on four ranges of particle sizes, with the smaller particles generally more reflective. (Data taken from the U.S. Geological Survey database [3].)

Here, the bidirectional reflectance r_{BRDF} is the ratio of the radiance $L(\theta_O, \phi_O)$ to the irradiance $E(\theta_i, \phi_i)$, where L is scattered into the direction described by the orientation angles θ_O and ϕ_O, and E is irradiance from the θ_i, ϕ_i direction (see Figure 2 for an illustration of BRDF geometry). The function in (1) describes the bidirectional reflectance values for every combination of input/output angles and, although not explicitly captured in the equation here, also changes as a function of wavelength. The simplest BRDF is constant with respect to the input and output angles; such a surface is termed *Lambertian*.

Although the assumption of an ideal diffusely reflecting surface may not truly be achieved in practice, it is often a useful approximation and a natural starting point for more sophisticated models. Some of these BRDF models are theoretical physics-based models that use simulations of light scattering from multifaceted and/or multilayered reflecting rough surfaces, while others are empirical models that provide simple formulations capable of reproducing specific kinds of reflective behaviors. Often, these models involve free parameters that are fit to comprehensive experimental measurements made by using sophisticated goniometers and gonioreflectometer systems; the challenge there is that those systems are only recently being designed for practical deployment in challenging environments [17], [18]. The more complex models are often used in computer graphic design to render realistic surface illustrations [19], and the simpler models are more commonly adopted in remote sensing. Schowengerdt [20] argues that most natural surfaces are approximately Lambertian for viewing angles (i.e., θ_O in Figure 2) within 20° to 40° from the zenith. But BRDF also depends on the illumination angle, with most surfaces brighter in the specular and backscatter directions and darker elsewhere. Trees and canopies, for instance, have a strong return from the backscatter direction [16]. Choosing the Lambertian model certainly simplifies hyperspectral data analysis, but it just as clearly underestimates the spectral variability that will be observed in both target and background materials.

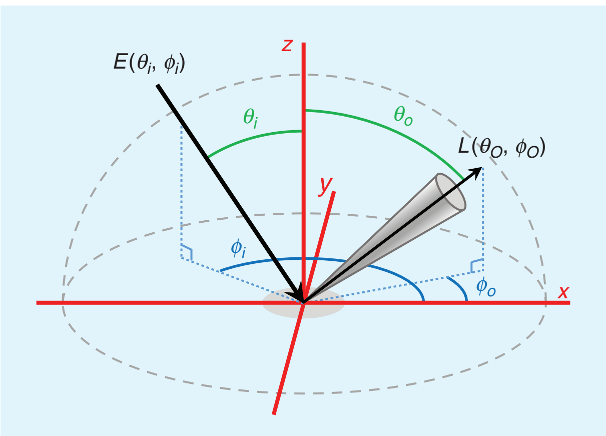

FIGURE 2. An illustration of the BRDF geometry described in (1).

Intimate mixtures present their own set of challenges, because their spectra result from complicated and varying underlying phenomenology (e.g., reflections and scatterings between grains of different materials). A common model for these nonlinear spectral interactions is Hapke's radiative transfer model for intimate mineral mixtures [21], [22]. This model considers several contributions to the spectra of intimate mixtures: single volume scattering, multiple volume scattering, coherent backscattering, the shadow hiding opposition effect, macroscopic roughness, and compactness. All of these contributions can lead to highly variable resulting spectra. A number of approaches have been explored for applications to hyperspectral unmixing, with particular focus on kernel-based methods [23]–[27].

EXTRINSIC TARGET VARIABILITY

The signal observed at the sensor depends not only on the nature of the material but also on how much material there is. For solid subpixel targets, the target size relative to the pixel size matters; for gas-phase plumes or thin layers of powder, concentration and thickness matter. In the LWIR, the temperature matters as well (i.e., warmer materials emit more radiation and appear brighter at the sensor).

Perhaps the simplest, and probably the most common, expression for how much material is in a pixel is given by the additive model [1]:

$$\mathbf{x} = \mathbf{z} + \epsilon \mathbf{t}, \tag{2}$$

where $\mathbf{x}$ is the observed spectrum at a pixel, $\mathbf{z}$ is the spectrum of the background material in that pixel, $\mathbf{t}$ is the spectral signature of the target material, and ϵ is a scalar quantity that corresponds to how much of the target is present.

For opaque subpixel targets, the larger the target, the more of the background it obscures, and a more appropriate expression is the replacement model [1]:

$$\mathbf{x} = (1-\alpha)\mathbf{z} + \alpha\mathbf{t}, \tag{3}$$

where $0 \le \alpha \le 1$ corresponds to the fraction of the pixel that is taken up by the target material.

Although the additive and replacement models are the most popular choices, the actual interaction of target and background can be more complicated. For chemical plumes, for instance, the additive model is popular, but it is only an approximation—accurate in the weak-plume limit [28]—of the actual Beer's law absorption rule [29]. This approximation is often valid, but it is also motivated by the fact that target detection algorithms are more easily derived under the assumptions of the additive model.

The scaled replacement model, which is a kind of hybrid of the additive and replacement models, considers a target that can both occlude the background and have variable strength:

$$\mathbf{x} = (1-\alpha)\mathbf{z} + \alpha\epsilon\mathbf{t}. \tag{4}$$

In the LWIR, this could, for instance, be due to temperature variation. An example in the VIS wavelengths arises when a target in the shade exhibits a lower apparent reflectance than that same target in the bright sun. More sophisticated models (e.g., those that reduce the dependence on the temperature [30] or achieve temperature-emissivity separation [31] or those that explicitly account for shadows [32]) may ultimately be preferable to the scaled replacement model, but they are also more complicated.

ENVIRONMENTALLY INDUCED TARGET VARIABILITY

Target spectral variability not only depends on the target itself but is also caused by the surrounding environment. In this subsection, we examine some of the most important environmental sources of spectral variability by considering the typical remote sensing scenario of a hyperspectral sensor mounted onboard a platform (such as an aircraft or a satellite) and imaging the targeted surface materials.

MAIN RADIATIVE TRANSFER MODEL EQUATIONS

The radiative transfer between a surface target and a sensor across the VIS and NIR (VNIR) (~400–1,000 nm) and SWIR (~1,000–2,500 nm) spectral ranges (i.e., VNIR–SWIR) can be expressed by the following simplified physics-based model for the sensor-reaching radiance $L^s(\lambda)$ [16], [33]–[36]:

$$L^s(\lambda) = L^{gr}(\lambda) + L^{sc}(\lambda) \tag{5}$$

$$L^{gr}(\lambda) = L^{dir}(\lambda) + L^{dif}(\lambda) + L^{obst}(\lambda) \tag{6}$$

$$L^{sc}(\lambda) = L^p(\lambda) + L^{adj}(\lambda). \tag{7}$$

A sketch of the various radiation contributions entering the sensor is provided in Figure 3, with Table 1 summarizing the major radiometric terms involved. In (5)–(7), the total ground-reflected spectral radiance $L^{gr}(\lambda)$ includes the contributions reflected by the target toward the sensor, and the solar scattered spectral radiance $L^{sc}(\lambda)$ includes the contributions scattered toward the sensor without having interacted with the targeted surface. $L^{dir}(\lambda)$ and $L^{dif}(\lambda)$ are the major contributors to $L^{gr}(\lambda)$ and represent, respectively, the direct and diffuse (downwelling) solar spectral radiances reflected by the target toward the sensor—also accounting for the multiple reflections between the adjacent material surface and the atmosphere due to multiple scattering [16]. The ground-reflected radiance also includes the spectral radiance $L^{obst}(\lambda)$ arriving on the target due to secondary illumination from potential nearby objects and obstacles in view of the target (such as trees and buildings) [35]. The solar scattered radiance $L^{sc}(\lambda)$ includes the so-called path radiance $L^p(\lambda)$—that is, the solar spectral radiance scattered by the atmosphere toward the sensor—and the adjacency term $L^{adj}(\lambda)$, accounting for the direct and diffuse solar spectral radiances reflected from the materials adjacent to the target toward the sensor (including the aforementioned multiple-reflections phenomenon).

By expanding the first two terms on the righthand side of (6), we obtain the following [16], [33], [36]:

$$\begin{aligned} L^{dir}(\lambda) + L^{dif}(\lambda) = \frac{E^{su}(\lambda) + F \cdot E^d(\lambda)}{1 - r_a(\lambda)S(\lambda)} \frac{r(\lambda)}{\pi} T^u(\lambda) \\ - \alpha_{sh} E^{su}(\lambda) \frac{r(\lambda)}{\pi} T^u(\lambda), \end{aligned} \tag{8}$$

$$E^{su}(\lambda) = E^{toa}(\lambda) \cos\theta \, T^d(\lambda). \tag{9}$$

In (8) and (9), $E^{su}(\lambda)$ is the solar unscattered (direct) irradiance incident on the surface, $E^{toa}(\lambda)$ is the solar exo-atmospheric spectral irradiance at the top of the atmosphere incident on a surface orthogonal to the sun's rays, θ is the sun zenith angle (subtended by the sun's rays and the surface normal), and $T^d(\lambda)$ is the downward atmospheric transmittance of the sun-to-surface path. The term $E^d(\lambda)$ is the downwelling spectral irradiance from the sky incident on the surface, which undergoes a scaling by the sky view factor (or shape factor) $F \in [0, 1]$ accounting for the fraction of sky dome visible from the target [35], [36]; $r(\lambda)$ is the diffuse spectral reflectance (or *albedo*) of the targeted surface [assumed to be Lambertian; i.e., $r(\lambda) = r_{BRDF}(\lambda) \cdot \pi$]; $r_a(\lambda)$ is the diffuse spectral reflectance of the material adjacent

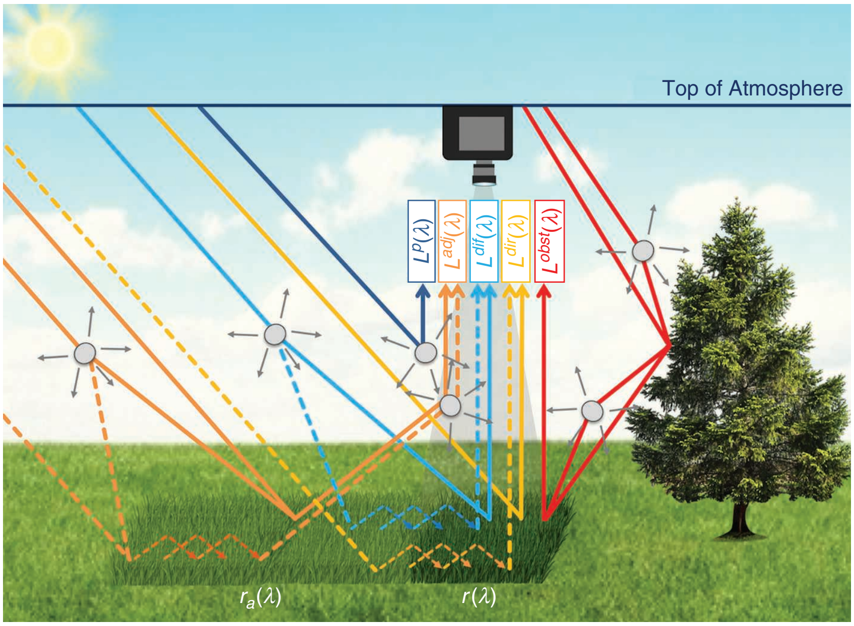

FIGURE 3. A sketch of the radiative-transfer model along the sun-to-target-to-sensor path, as described in the "Main Radiative-Transfer Model Equations" section. The individual terms are listed in Table 1.

to the target; $S(\lambda)$ is the atmospheric spherical scattering albedo (i.e., the effective diffuse reflectivity of the sky to upwelling radiation); and $T^u(\lambda)$ is the upward atmospheric transmittance of the target-to-sensor path. The denominator in the first term of (8) is due to the multiple reflections between the adjacent material surface and the atmosphere due to multiple scattering. This phenomenon, also known as the *trapping effect*, has been modeled for a flat and homogeneous surface as the summation of infinite terms of reflected contributions, leading to a geometric series that converges to $[1-r_a(\lambda)S(\lambda)]^{-1}$. This effect has been found negligible for clear-sky conditions and low reflectivity of adjacent materials, i.e., $r_a(\lambda)S(\lambda) < 0.02$ [16]. The second term in (8) accounts for potential shadowing of the direct solar unscattered term, with $\alpha_{sh} = 1$ for full shadowing and $\alpha_{sh} = 0$ for full illumination [33].

For spectral remote sensing, the factor we care most about in these equations is $r(\lambda)$, the reflectance. This is what corresponds to the intrinsic properties of the material on the ground, that is, the material we are trying to find, identify, or characterize. Extracting $r(\lambda)$ from the sensor-reaching radiance $L^s(\lambda)$ is no small feat, because, in doing so, we not only have to estimate all of the other terms and factors in (5)–(9) but also have to deal with the variability of each of those terms and factors.

The terms $L^{obst}(\lambda)$ and $L^{adj}(\lambda)$ in (6) and (7) are not expanded here, but details can be found in [16], [35], [37], and [38]. $L^{adj}(\lambda)$ has basically the same form as (8), with $r_a(\lambda)$ in place of $r(\lambda)$ and with the diffuse upward transmittance instead of $T^u(\lambda)$. As for $L^{obst}(\lambda)$, this has been modeled by accounting for the presence of a nearby object that obstructs part of the sky dome viewable from the target; thus, the average spectral radiance reflected by the object itself is scaled by a factor $(1-F)$ [16], [35]. Depending on the relative position of the sun, target, and obstacle, $L^{obst}(\lambda)$ may also include the solar direct radiation reflected by the nearby object toward the target.

TABLE 1. SUMMARY OF THE RADIOMETRIC TERMS INVOLVED IN RADIATIVE TRANSFER MODELING.

TERM	SPECTRAL RADIANCE
$L^s(\lambda)$	Sensor-reaching spectral radiance
$L^{gr}(\lambda)$	Total ground-reflected spectral radiance
$L^{sc}(\lambda)$	Solar scattered spectral radiance
$L^{dir}(\lambda)$	Direct solar spectral radiance
$L^{dif}(\lambda)$	Diffuse (downwelling) solar-spectral radiance
$L^{obst}(\lambda)$	Nearby obstacles/objects spectral radiance (secondary illumination)
$L^p(\lambda)$	Solar path-scattered radiance (path radiance)
$L^{adj}(\lambda)$	Adjacency spectral radiance
	SPECTRAL IRRADIANCE
$E^{toa}(\lambda)$	Exo-atmospheric spectral irradiance at the top of the atmosphere
$E^{su}(\lambda)$	Solar unscattered (direct) spectral irradiance incident on the surface
$E^d(\lambda)$	Downwelling spectral irradiance from the sky incident on the surface
$E^{su}(\lambda; x, y)$	Solar unscattered spectral irradiance incident on a tilted surface at (x, y)
$E^d(\lambda; x, y)$	Downwelling spectral irradiance incident on a tilted surface at (x, y)
	SPECTRAL TRANSMITTANCE
$T^u(\lambda)$	Upward atmospheric transmittance (target-to-sensor path)
$T^d(\lambda)$	Downward atmospheric transmittance (sun-to-surface path)
	SPECTRAL REFLECTANCE
$r(\lambda)$	Spectral reflectance of the target (assumed Lambertian)
$r_a(\lambda)$	Spectral reflectance of the adjacent material
$S(\lambda)$	Atmospheric spherical-scattering albedo

TOPOGRAPHIC EFFECTS

So far, we have assumed a flat surface, but topographic effects and their possible incorporation into the model have been thoroughly studied (see [34], [37], and [39]–[42]), and (8) and (9) can be modified accordingly. Specifically, for a local target surface at location (x, y) tilted by a slope angle $\theta_{sl}(x,y)$ (subtended by the surface normal and the vertical direction), (8) and (9) should be modified by accounting for the local direct and diffuse irradiances at (x, y):

$$E^{su}(\lambda; x, y) = E^{toa}(\lambda)\cos[\theta_{\mathrm{ill}}(x,y)]T^d(\lambda) \tag{10}$$

$$F(x,y)\cdot E^d(\lambda; x, y) = E^d(\lambda)\cdot\left\{A(\lambda)\frac{\cos[\theta_{\mathrm{ill}}(x,y)]}{\cos\theta} + [1-A(\lambda)]F(x,y)\right\} \tag{11}$$

$$\cos[\theta_{\mathrm{ill}}(x,y)] = \cos\theta\cos[\theta_{sl}(x,y)] + \sin\theta\sin[\theta_{sl}(x,y)]\cos[\phi-\phi_{sl}(x,y)]. \tag{12}$$

Here, $\phi_{sl}(x, y)$ is the azimuth angle of the local sloped terrain at (x, y), ϕ is the sun azimuth angle, and $\theta_{ill}(x,y)$ is the local (actual) target illumination angle. In (11), the diffuse component is expressed as a linear combination of two terms: one circumsolar diffuse irradiance from the solid angle near the sun and one isotropic contribution from the remaining sky dome [37], [39], [40]. $A(\lambda)$ is the anisotropy index, which gives the proportion of diffuse radiation to be treated as circumsolar and isotropic and has been modeled as $A(\lambda) = (1-\alpha_{sh})T^d(\lambda)$ [37], [39], [40]. The sky-view factor for the tilted target surface in (11) should be evaluated from digital elevation models [40], [43], although simplified expressions have also been derived (e.g., for an infinitely long slope [40]). Of note is that (10)–(12) can be modified or enriched to account for topography at different levels, such as for mountainous terrain [37], or for the combined effect of topography and structured forests [41], [42]. Additionally, some works use a sky radiance fraction function instead of scaling the downwelling irradiance by the sky-view factor in (8) and (11), in turn obtaining a wavelength- and angle-dependent term accounting for the non-Lambertian nature of the sky reflectance [44].

As shown in these equations, many factors are involved in the radiative transfer process. Each of these factors affects the spectral intensity and shape of the sensor-measured radiance spectrum and can ultimately result in a significant amount of spectral variability. The sources of variability may be broadly categorized into four areas: atmosphere, illumination, acquisition geometry, and adjacent environment. In the following section, we examine the effects of this variability through a detailed analysis based on the radiative transfer model equations just described.

ANALYSIS OF ENVIRONMENTALLY INDUCED VARIABILITY

Most of the variability in an observed spectrum is due to atmosphere and illumination, and these sources of variability have indeed been the most widely investigated [36], [45]–[48]. Many have shown, either by simulation or by field experiments, how much the spectral radiance for a single material can vary under different atmospheric conditions [e.g., aerosol/gas types and concentrations or water vapor (WV) profiles] [46] and illumination conditions (e.g., shadowing or sky-dome obstruction) [36], [45], [47], [48]. To investigate adjacency effects and secondary illumination, Goa et al. [48] looked at radiance measurements of targets placed between tree lines and exposed to different illumination conditions; these measurements were then compared with synthetic radiance spectra simulated according to a model based on (5)–(7). The results revealed the importance of accounting for the secondary illumination term $L^{obst}(\lambda)$, which is often neglected, and showed that it can play an important role in cases where, for instance, the target is in shadow and surrounded by tall and brightly lit objects [35]. Similar considerations can be made for the effects of adjacent materials in terms of both $L^{adj}(\lambda)$ and $[1 - r_a(\lambda)S(\lambda)]$. They may not be negligible, especially when $r_a(\lambda) \gg r(\lambda)$ [37], that is, when adjacent materials are more reflective than target materials (such as targets surrounded by brighter backgrounds and, in particular, targets in shadow) [35], [49]. Viewing conditions and acquisition geometry also play important roles in the spectral variability of materials [46], [50], especially topographic effects that may considerably increase or decrease the direct and diffuse illumination of the target. For example, consider a sun zenith angle of 25°, a slope angle of the target surface of 15° away from the sun, and a null azimuthal difference between the sun and the slope orientation. In this case, the direct illumination would decrease by approximately 15% compared with the flat-terrain case. For the same geometry but with a sun zenith angle of 45°, the decrease would be approximately 30%.

The atmospheric effects of transmittance and visibility in general tend to vary slowly with respect to spatial position and are often assumed to be stationary over a full scene. This assumption can fail for wide-area remote sensing or with shallow slant-angled views (called *oblique sensing*) [50], [51], where the target-to-sensor path distance may vary considerably over a scene. Other sources of variability, such as columnar WV [52], clouds, shadows, and obstacles (e.g., leafy canopies and buildings), can be quite spatially nonstationary. Topographic effects also exhibit a strong spatial nonstationarity within a scene. In part, this is because of the increased range of the sensor viewing angle within a scene [50], [51], but a further effect of oblique sensing is that surfaces that were previously hidden become visible (e.g., walls of a building as well as the roof or trunks of trees as well as the canopy) [51]. Adjacency effects are intrinsically nonstationary, of course, because they depend on the spatial variations of surface materials in the examined scene.

A summary of these considerations is given in Table 2. Graphical examples showing how these sources of variability manifest themselves on the sensor-reaching radiance are provided in the following section.

EXAMPLES OF ENVIRONMENTALLY INDUCED VARIABILITY EFFECTS

An example experiment is shown in Figure 4, where at-sensor radiance spectra in the VNIR were synthetically generated by exploiting spectral signatures from the ASTER spectral library [53]. The spectral reflectance $r_{Cu}(\lambda)$ of copper was used for the target material. This was chosen because it has a smooth reflectance spectrum and exhibits both low reflectivity (in the blue/green portions of the VIS) and medium/high reflectivity (in the red and NIR). To illustrate the effects of adjacent and obstacle materials with respect to the copper, we used a rangeland reflectance spectrum $r_{Rl}(\lambda)$, with very low reflectivity across the full VNIR; an olive canopy reflectance spectrum $r_{Ol}(\lambda)$, with high reflectivity in the NIR only; and an aluminum reflectance spectrum $r_{Al}(\lambda)$, with higher reflectivity across the full VNIR. A plot of these four spectra is shown in Figure 4(a).

In this example, we used the MODTRAN (i.e., MODerate resolution atmospheric TRANsmission) 5 radiative transfer code [54] to simulate radiation transfer in the atmosphere in combination with the physical models expressed by (10)–(12). A typical airborne remote sensing scenario was reproduced by making reference to a real hyperspectral data collection campaign performed in May 2013 with the SIM.GA hyperspectral sensor (511 spectral channels in the VNIR with a full width at half maximum of about 2 nm), flying over the city of Viareggio, Italy [55]. The parameters are listed in Table 3, where the values of the varied parameters are reported in curly brackets. One single parameter was varied each time by keeping the others fixed at the basic configuration value highlighted in boldface. In Figure 4(b)–(h), the spectra corresponding to the basic configurations are displayed in blue in each plot. Each plot legend shows how the other colors are assigned to the variations.

Specifically, we varied the horizontal visibility (the surface meteorological-range VIS parameter in MODTRAN) in Figure 4(b); the WV scaling factor (H2OSTR in MODTRAN) in Figure 4(c); the sensor viewing angle θ_v (ANGLE in MODTRAN) in Figure 4(d); shadowing (by means of α_{sh}) in Figure 4(e); topography (by taking $\phi - \phi_{sl} = 0$ and varying θ_{sl}) in Figure 4(f); adjacency effects (using a null-albedo or olive rangeland or aluminum signature for the adjacent material) in Figure 4(g); and obstacles and obstruction (using a null-albedo or olive rangeland or aluminum signature for both 15% and 30% sky-dome obstructions) in Figure 4(h).

With respect to the basic configuration of VIS = 23 km (the typical default value in MODTRAN for good visibility conditions), visibility variations strongly affect the spectra, as shown in Figure 4(b). The major effects can be observed in the low-wavelength region for low VIS values, where the low target reflectivity combined with the increased scattering phenomena due to the hazier conditions results in considerable increases of the at-sensor spectral radiance. For higher wavelengths, where scattering not only is generally weaker but the copper target also has higher reflectance, the general effect on the at-sensor radiance is a decrease of the signal for low VIS values due to the less transparent atmosphere (lower transmittance). The result is a completely distorted signal. In contrast, in Figure 4(c), the effects of WV can be seen to be mostly confined to the WV absorption bands, with decreased signal (more absorption) for more humid atmospheres (higher WV scaling). Oblique sensing [Figure 4(d)] plays a significant role only for very slanted views (e.g., $\theta_v = 120°$), which mostly affect lower wavelengths with an increased at-sensor radiance due to the increased scattering phenomena occurring in the longer target-to-sensor atmospheric path. Very oblique views also affect, although to a lesser extent, the signal in the major absorption bands (e.g., at 940 nm), where decreased signal intensities can be observed due to the higher absorption. The reduction of the direct solar illumination due to shadowing entails considerable variability over the at-sensor spectral radiance in Figure 4(e), with a general decrease of the at-sensor signal throughout the VNIR range. Similar variations can be observed in Figure 4(f), where varying the inclination of the targeted surface determines a general increase (if the surface is rotated toward the sun) or decrease (if it is rotated away from the sun) of the at-sensor signal. In the simulated scenario, the sun zenith angle is $\theta = \theta_i \approx 31°$, and thus an inclination of $\theta_{sl} = 30°$ is close to the condition of maximum illumination due to topography.

The adjacency effects shown in Figure 4(g) illustrate how much those effects depend on the properties of the adjacent material itself. In this example, the low-reflectivity rangeland spectrum hardly affects the resulting at-sensor radiance. Conversely, for adjacent materials with stronger reflection properties, such as the olive or the aluminum materials, adjacent effects are more significant. The olive signature "colors" the at-sensor radiance according to its own spectral characteristics, with adjacency effects mostly manifesting themselves in the NIR spectral range; in the VIS, these effects are negligible, because the olive material and copper have a null contrast in that range of the spectrum. The aluminum object, which has a higher reflectivity than copper throughout the spectral range, causes a general increase in the at-sensor signal. Similar considerations can be found for the secondary illumination due to nearby objects [see Figure 4(h)], which cannot be considered negligible for high-reflectivity objects. Here, the shape factor induces

TABLE 2. A CATEGORIZATION OF ENVIRONMENTALLY INDUCED SOURCES OF VARIABILITY.

	SOURCE OF VARIABILITY	MAINLY AFFECTED TERMS	SPATIAL STATIONARITY	TEMPORAL STATIONARITY	
	Constituents, gases, aerosols	$E^d(\lambda)$, $L^p(\lambda)$, $T^d(\lambda)$, $T^u(\lambda)$, $S(\lambda)$	Stationary over wide areas*	Seasonal	ATMOSPHERE
	Clouds, WV	$T^d(\lambda)$, $T^u(\lambda)$, $L^p(\lambda)$, $E^d(\lambda)$, $S(\lambda)$	Nonstationary	Nonstationary	ATMOSPHERE
ADJACENT ENVIRONMENT	Adjacent materials	$L^{adj}(\lambda)$, $r^a(\lambda)$	Nonstationary	Stationary	
ADJACENT ENVIRONMENT	Nearby objects and obstacles	$L^{dif}(\lambda)$, $L^{obst}(\lambda)$	Nonstationary	Stationary	ILLUMINATION
ADJACENT ENVIRONMENT	Shadows	$L^{dir}(\lambda)$	Nonstationary	Hourly variations	ILLUMINATION
ADJACENT ENVIRONMENT	Sensor viewing angle	$T^u(\lambda)$, $L^p(\lambda)$	Stationary**	Stationary	ILLUMINATION
ACQUISITION GEOMETRY	Sun position	$E^{su}(\lambda)$, $E^d(\lambda)$, $L^{obst}(\lambda)$, $L^{adj}(\lambda)$, $T^d(\lambda)$, $T^u(\lambda)$	Stationary over wide areas	Hourly variations	ILLUMINATION
ACQUISITION GEOMETRY	Topography	$E^{su}(\lambda)$, $L^{dif}(\lambda)$, $L^{obst}(\lambda)$, $L^{adj}(\lambda)$	Nonstationary	Stationary	

*Except for wide-imaged areas and situations with extremely wide-angle views or very shallow slant-angled views.
**Except for situations with extremely wide-angle views.

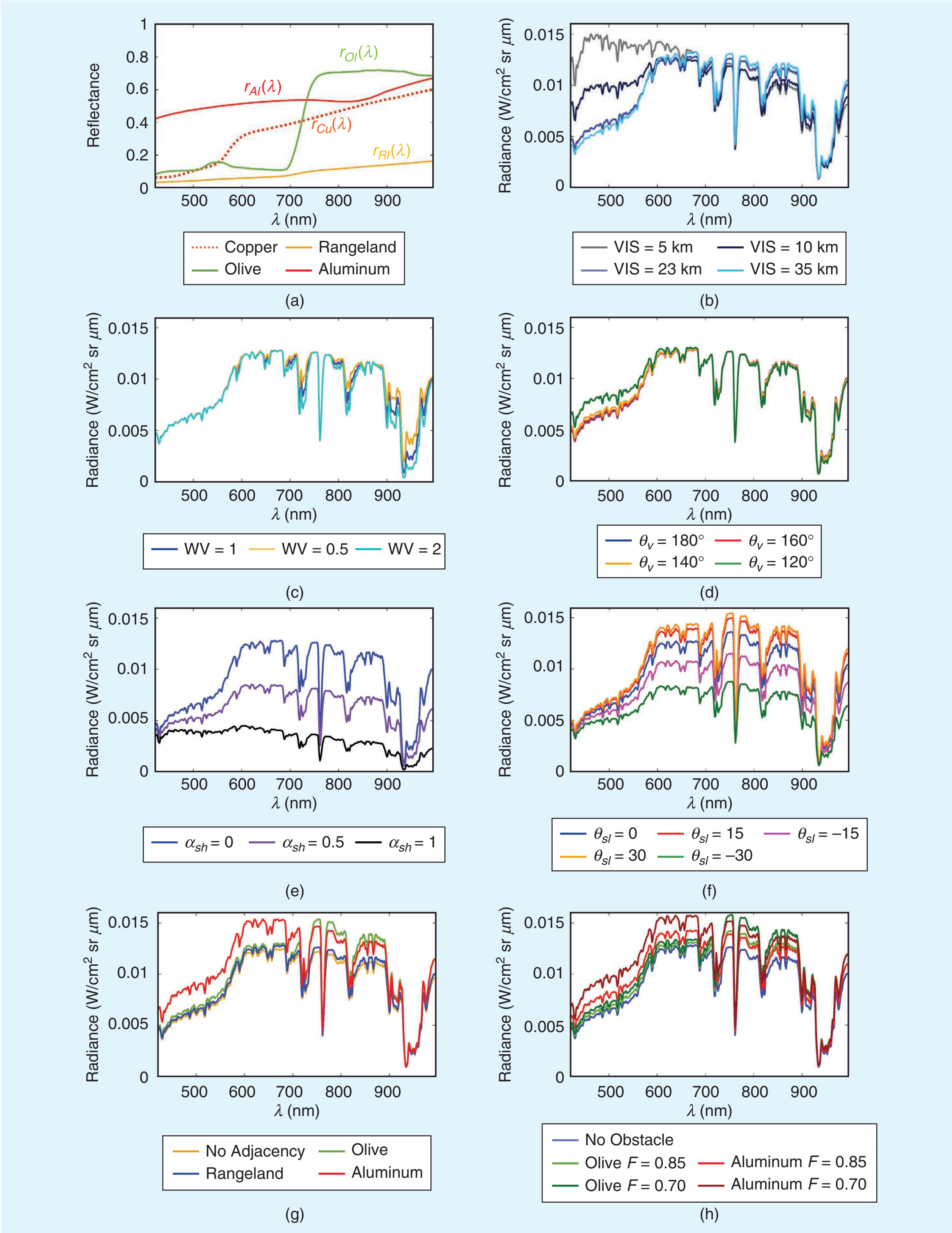

FIGURE 4. An example showing the major effects of environmental variability sources over the spectrum of a copper object. (a) The spectral signatures used. The dotted signature $r_{Cu}(\lambda)$ is the reflectance of the copper object that served as target. The other signatures were used to simulate adjacency effects and secondary illumination by obstructing obstacles. (b) The effects of visibility (VIS) and (c) the effects of the WV scaling factor (WV_{sf}). (d) The effect of the sensor viewing angle, measured from the zenith ($\theta_v = 180^\circ - \theta_0$; note that $\theta_v = 180^\circ$ corresponds to the nadir view). (e) The shadowing, (f) topographic, and (g) adjacency effects and (h) the effects due to secondary illumination by obstacles.

variability as well, with an increasing secondary illumination impact for higher obstruction levels.

Although BRDF effects are intrinsic to the target material (and have been described in the "Intrinsic Target Variability" section), spectral variability due to BRDF also has an environmental flavor. Indeed, variability due to variations in illumination and acquisition geometry is deeply intertwined with variability due to the non-Lambertian nature of target materials. Thus, to provide some insight into environmentally induced spectral variability for non-Lambertian targets, a further illustrative example, using the same parameters as used previously, is shown in Figure 5. Two glossy targets taken from the Cornell BRDF database [56] are considered here: specifically, a target painted with a "garnet red" lacquer [Figure 5(a) and (b), top panels] and a glossier target painted with a "mystique" lacquer [Figure 5(c) and (d), bottom panels]. Here, the left panels in Figure 5(a) and (c) plot the target BRDFs for a fixed illumination direction (as before, $\theta_i = \theta_i \approx 31^\circ$) for varying orientation angle θ_O. The values of θ_O were chosen to correspond to the same sensor-viewing angles, $\theta_v = 180^\circ - \theta_O$, used in the previous example. The oblique specular geometry ($\theta_v = 149^\circ$) is not included in the analysis, because the Cornell BRDF data do not include measurements of specular reflectance; if needed, however, they could be introduced by adding a term ruled by the Fresnel reflectance [57].

As is evident from the target BRDFs, the non-Lambertian effects are different for the different targets. In Figure 5(a), the effect is mostly just a scaling of the reflectance function for the "garnet red" target, with higher reflectance for nadir viewing. In Figure 5(c), however, the effect produces actual changes of color for the "mystique" target, which exhibits a strong reflectance contribution of approximately 500 nm for nadir and 20° off-nadir views and strong contributions at shorter and higher wavelengths for the 40° and 60° off-nadir views. Figure 5(b) and (d) plots the sensor-reaching radiance for the sensor-viewing angles matching the orientation angles varied in Figure 5(a) and (c). The figures clearly show that environmental effects blend with the BRDF effects in the at-sensor spectral radiances. In fact, whereas the mostly path-radiance-driven effects already observed in Figure 4(d) (i.e., increased radiance, especially at shorter wavelengths for longer paths due to oblique sensing) are clearly evident (and stronger with respect to the previous examples due to the very low reflectance of the targets), these are combined with the effects of the different spectral characteristics that the target materials intrinsically have for different viewing angles.

In Figure 5(b), the path-radiance-driven spectral radiance increase at shorter wavelengths, strongly evident for the most oblique case ($\theta_v = 120^\circ$), is less evident for the other off-nadir viewing angles because the higher overall reflectance for a nadir view [evident in Figure 5(a)] determines a stronger radiance contribution, even though the increase due to path radiance is lower than in the off-nadir cases. Similar considerations can be made in Figure 5(d), where the stronger reflectance contributions around 500 nm for the "mystique" target at nadir and 20° off-nadir views lead to radiance contributions around 500 nm that are stronger than or equal to the 40° off-nadir case.

These simple examples demonstrate that the causes of nonnegligible spectral variability are multifold, and the corresponding effects on the target spectra manifest themselves in complicated ways that can go beyond straightforward scaling or offsets. The aggregation of multiple diverse environmental effects—in addition to the spatial nonstationarity and temporal nonstationarity of the sources of these effects—makes environmentally induced spectral variability a true challenge for effective hyperspectral image exploitation.

So far, we have examined the major sources of environmentally induced variability occurring when spectral radiance is measured in the VNIR–SWIR range. If data collection takes place with a sensor operating in the LWIR, a different radiative transfer model should be considered [58], [59], and a further source of variability comes into play, namely, temperature (primarily of the surface, but also of the atmosphere and the adjacent areas), which exhibits further temporal and spatial nonstationarity.

We conclude this section, which focused on the main sources of variability for the sensor-reaching spectral radiance $L^s(\lambda)$, with a reminder that $L^s(\lambda)$ as expressed here is the radiance entering the sensor, which is not precisely equal to the spectral radiance actually measured by the sensor. Although beyond the scope of this survey, radiance collection by the sensor acquisition system can introduce further variability to the measured signal (e.g., due to sensor noise, smile and keystone effects,

TABLE 3. PARAMETERS USED IN THE EXAMPLE IN FIGURE 4.

ATMOSPHERIC MODEL	Midlatitude summer	**VISIBILITY (km)**	{5, 10, **23**, 35}
AEROSOL MODEL	Maritime	**WV SCALING FACTOR (WV_{sf})**	{0.5, **1**, 2}
SENSOR ELEVATION (km)	1.250	**ADJACENCY**	$r_a = \{\mathbf{0}, r_{OI}, r_{RI}, r_{AI}\}$
TARGET ALTITUDE (km)	0	**OBSTACLES AND OBSTRUCTION**	$r_{\mathrm{obst}} = \{\mathbf{0}, r_{OI}, r_{AI}\}$ $F = \{\mathbf{0}, 0.85, 0.70\}$
TARGET LATITUDE (°)	43.85 north	**VIEWING ANGLE (θ_v) [°]**	{120, 140, 160, **180**}
TARGET LONGITUDE (°)	10.25 east	**SHADOWING (α_{sh})**	{0, 0.5, **1**}
DATE	9 May 2013	**SLOPE ELEVATION (θ_{sl}) [°]**	{**0**, ±15, ±30}
GMT TIME (h)	12.65	**SLOPE/SUN RELATIVE AZIMUTH ($\phi_{sl} - \phi$) [°]**	0

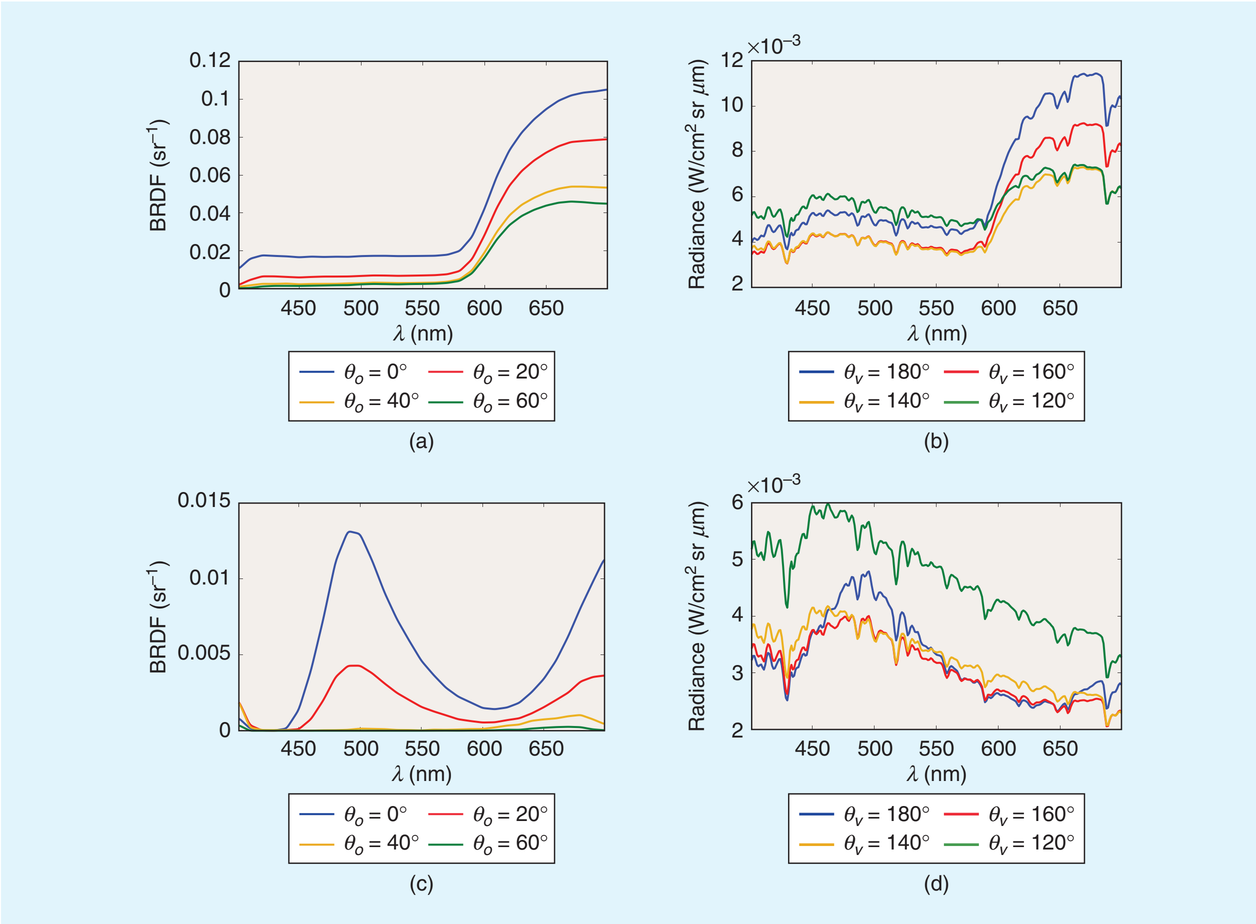

FIGURE 5. An example showing the effects of sensor viewing-angle variation for two non-Lambertian targets. (a) and (b) The target painted with "garnet red" lacquer. (c) and (d) The target painted with the glossier "mystique" lacquer. (a) and (c) The BRDF of the targets plotted for a fixed illumination angle ($\theta_i = \theta$) and different orientation angles θ_O. In the "garnet red" target, BRDF effects mostly consisted of scaling of the reflectance functions with varying orientation angles, whereas the "mystique" target actually changes color depending on θ_O. (b) and (d) The corresponding at-sensor radiances for sensor viewing angles $\theta_v = 180^\circ - \theta_O$ matching the orientation angles shown in (a) and (c).

pixel nonuniformity, point spread function, and lens distortions) [1], [60].

MODELS FOR SPECTRAL VARIABILITY

In attempting to capture the complex spectral variability observed in target materials, most models fall into two classes: statistical methods, which involve fitting a probability distribution function (pdf) to the variability, and geometrical methods, which invoke structures such as subspaces, simplexes, and manifolds. Some of these are sketched in Figure 6, and they are described in a bit more detail in the following sections.

PROBABILISTIC MODELS

Never tell me the odds.

—Han Solo

Spectral variability amounts to the same thing as spectral uncertainty, and the most natural tool for characterizing uncertainty is the pdf. When *ab initio* physical models are lacking, generic distributions are appealing, and, for characterizing complex backgrounds, probabilistic models are the dominant choice [61]. For characterizing the spectral variability of specific target materials, however, we often have more physical insight (e.g., as detailed in the "Sources of Target Spectral Variability" section). Incorporating that physical insight into algorithms is often a challenge, however, and probabilistic models are widely used. The Gaussian pdf has many advantages [62], and several authors [63], [64] have considered a Gaussian target variability that matches (except for offset and scale) the covariance of the background variability. There may not be a strong physical argument to expect the target and background covariances to line up in this way, but it simplifies the analysis (whitening the background has the effect of whitening the target as well), and it is at least more realistic than assuming the target has zero covariance. In other work that uses the Gaussian distribution to model the spectral variability of different target materials, each specific material was considered with its own covariance matrix [65].

Tyo et al. [66] evaluated several monovariate distributions besides Gaussian for modeling both the single-band spectral radiance and the total radiance (summed across all bands) extracted from multiple instances of the same object in a hyperspectral image. They observed that the extracted data exhibited a skewed distribution and, therefore, used log-normal, gamma, and Weibull pdfs for fitting. Experiments showed that none of the tested pdfs did a very good job in accurately modeling the statistical variability of the spectral signature (with a slight advantage for the log-normal distribution when the total radiance was modeled) and suggested that other more complex distributions should be tested. In [67], the univariate beta distribution was found effective at modeling the spectral variability affecting target material reflectances at each spectral dimension while incorporating distributional skew and ensuring that the reflectance values of the target materials are constrained to a physically realistic range. To approximate any distribution that the target materials may exhibit, such as the multimodal distributions very commonly found in real data, a mixture of Gaussian distributions was used in [68]. These models based on beta and a mixture of Gaussian distributions were specifically used in unmixing applications in the context of endmember spectral variability [67], [68]. Simplex-based distributions have also been suggested for anomaly detection [69].

UNCONSTRAINED SUBSPACE MODELS

Subspaces have been widely utilized to model the spectral variability of observed material spectra [36], [45], [46], [70], [71]. With a subspace model, the target material vector is restricted to vary in a K-dimensional subspace of the d-dimensional data space (with $K \leq d$). The amount of variability allowed increases as K increases from $K = 1$ (scalar variability in one dimension) to $K = d$ (the spectral vector is allowed to vary over all of $\mathbb{R}^d$). This can be written as

$$\mathbf{t} = \sum_{k=1}^{K} a_k \mathbf{t}_k = \mathbf{Ta}, \tag{13}$$

where $\mathbf{t} \in \mathbb{R}^d$ is the material spectrum, $\mathbf{T}$ is the $d \times K$ target subspace matrix, and $\mathbf{a} \in \mathbb{R}^K$ is a vector of coefficients. According to (13), the target material spectrum is expressed as a linear combination of the subspace basis vectors $\{\mathbf{t}_k\}_{k=1}^{K}$ weighted by the elements of the coefficient vector $\mathbf{a}$. If $rank\{\mathbf{T}\} = K$, then the basis vectors are linearly independent, and a given $\mathbf{t}$ will uniquely define the coefficient a_k.

The affine model [72] is a useful extension to the subspace model that is obtained by adding a nonzero offset:

$$\mathbf{t} = \mathbf{t}_o + \mathbf{Ta}. \tag{14}$$

One scenario where this arises is when the coefficient $\mathbf{a}$ corresponds to the amount of direct solar illumination and $\mathbf{t}_o$ corresponds to the spectrum of the target in the shade.

CONSTRAINED SUBSPACE MODELS

In (13), the coefficients in vector $\mathbf{a}$ are unconstrained. If they are to be interpreted as physical quantities (such as abundances), however, then constraints can be used to produce more realistic models of target variability. In particular, nonnegativity and sum-to-one constraints for the $\{a_k\}_{k=1}^{K}$ coefficients can be imposed, thus leading to a simplex model:

$$\begin{cases} \mathbf{t} = \sum_{k=1}^{K} a_k \mathbf{e}_k = \mathbf{Ea} \\ \mathbf{a} \geq 0 \\ \mathbf{1}^{\top}\mathbf{a} = 1 \end{cases}, \tag{15}$$

where $\mathbf{1}$ is a K-dimensional column vector including all ones, the superscript $\top$ denotes the transpose operator, and the columns $\mathbf{e}_k$ of the $d \times K$ matrix $\mathbf{E}$ can be interpreted as the K vertices (or endmembers) of the $K-1$ dimensional simplex, inside which the target $\mathbf{t}$ is constrained to lie. This is essentially identical to the linear mixing model, first proposed by Boardman [73] and now very widely used for background modeling [61], [74]. If the target is being used in an additive model, then the overall magnitude is not important, and

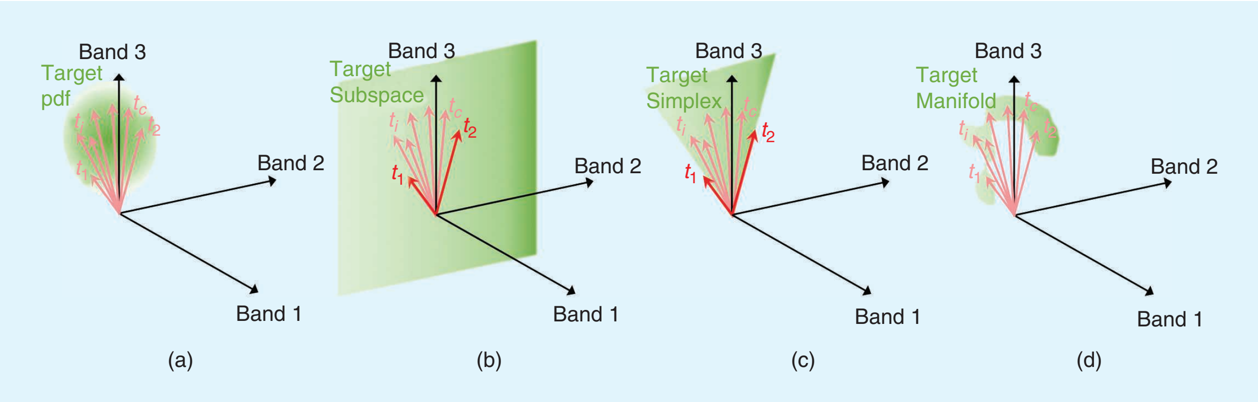

FIGURE 6. A sketch of the target variability models: (a) probabilistic, in which the target vectors are drawn from a probability distribution on $\mathbf{t}$; (b) subspace, in which the target vectors are linear combinations of $\mathbf{t}_1$ and $\mathbf{t}_2$; (c) constrained subspace, in which the target vectors are positive linear combinations of $\mathbf{t}_1$ and $\mathbf{t}_2$; and (d) manifold, in which the target vectors lie in a potentially intricate and convoluted structure that locally appears to be Euclidean.

there is no reason to enforce the sum-to-one constraint. In this case, the constraining geometry becomes an uncapped simplex (i.e., a polyhedral cone with K edges) [71].

With constraints imposed, it is not strictly necessary that the target subspace be of lower dimension than the ambient dimension of the data. The simple geometric approach adopted by Yang et al. [75] defines a ball $\mathcal{B}(\mathbf{t}_o, \varepsilon) = \{\mathbf{t} \mid \|\mathbf{t} - \mathbf{t}_o\| \leq \varepsilon\}$ centered on a nominal target spectrum $\mathbf{t}_O$ and having a radius ε corresponding to the variability of the target. The target spectrum is assumed to lie within the ball, and, when a detection algorithm is derived, the optimization is constrained by $\mathbf{t} \in \mathcal{B}$.

A different way to constrain a subspace model is by imposing a sparsity constraint on the coefficients. The model becomes a union of lower-dimensional subspaces and has been used to produce target-detection algorithms (e.g., [76]).

TOPOLOGICAL MANIFOLDS

Although the variability of material spectra can sometimes be modeled well by low-dimensional linear subspaces, the inherent nonlinearities of many physical processes that lead to this variability (e.g., those discussed in the "Sources of Target Spectral Variability" section) suggest that a more suitable model might be a "curved subspace" or nonlinear manifold [77], [78]. Manifolds can be linear or nonlinear, and *manifold learning* refers to approaches that attempt to recover (i.e., "learn") a lower-dimensional manifold assumed to be embedded in a higher-dimensional space [79]. Although hyperspectral data are more commonly modeled with linear manifolds (i.e., constrained and unconstrained subspaces), some hyperspectral data sets have been shown to be more effectively fit with nonlinear manifolds [80], [81].

Within this framework, nonlinear manifold learning attempts to derive a coordinate system that parameterizes the manifold by, in the words of Bachmann et al. [77], "following its intricate and convoluted structure with the hope of achieving a better data representation." In nonlinear manifold learning, the concept of linear distance is replaced by that of geodesic distance [82], which corresponds to the length of the shortest path on the manifold between two data points. In practice, this means that the manifold coordinate system resides on the manifold itself, so that the distances are measured by following the curves of the manifold trajectory; thus, any geodesic distance along the manifold turns out to be a simple linear distance in the manifold coordinate system [77]. Formally, an m-dimensional manifold $\mathcal{M}$ is defined by stating that, for each point $\mathbf{y} \in \mathcal{M}$, there is an open subset (often called a *neighborhood*) $S \subset \mathcal{M}$, with $\mathbf{y} \in S$, that is homeomorphic to an open subset (or neighborhood) S' in an m-dimensional Euclidean space $\mathbb{R}^m$; that is, $S \xrightarrow{g} S'$, where g is the homeomorphic mapping function. Thus, even though a manifold might have a complex nonlinear global structure, locally it looks like Euclidean space [77].

Well-known manifold-learning algorithms include kernel principal component analysis [83], isometric mapping [84], locally linear embedding [85], [86], and Laplacian eigenmaps [87]. They have been applied to a variety of hyperspectral image exploitation tasks [77], [80], [81], [88]–[95], but manifold learning in hyperspectral imaging is still a growing research topic [78].

MANAGING SPECTRAL VARIABILITY IN HYPERSPECTRAL ANALYSIS

Stay on target.

—Gold Five

Spectral variability takes many forms, and there have been many approaches for dealing with this variability in hyperspectral data analysis. We divide those strategies into two categories. The first, which we explore in the "Developing Algorithms That Are Robust to Spectral Variability" section, is to design algorithms that are robust to this variability. In this section, we emphasize algorithms for target detection, but we note that classification, segmentation, and unmixing tasks also require attention to spectral variability of materials. The second kind of strategy is to preprocess the data to reduce (in some cases, to "project out") the variability due to factors that are not part of our analysis. In the "Mitigating Spectral Variability in Hyperspectral Data" section, we describe both physics-based approaches, such as atmospheric compensation (AC), and data-driven approaches, such as in-scene target characterization.

Modeling the spectral variability of materials is particularly important for target detection. Much of the target-detection literature has been focused on how to incorporate variability of the background into the detection algorithms [61], but the variability of the target material must be accounted for as well. Even though variability effects for the target material may be milder than those of the background, if not properly accounted for, they will unavoidably lead to performance degradation. One reason target variability is more difficult to model is the scarceness of labeled training samples, which often consist of a single library spectrum for each material. When, instead, a greater number of labeled training samples is available (as in classification), these may be representative, albeit partially, of the spectral variability to be experienced in the scene. Another difficulty with target detection is the need to transform between the reflectance and radiance domains (spectral libraries are in reflectance; remote sensing measurements are in radiance), and this leads to a host of environmental sources (e.g., atmospheric absorption and scattering, angle of illumination, clouds, shadows, and so on) of variability.

DEVELOPING ALGORITHMS THAT ARE ROBUST TO SPECTRAL VARIABILITY

The most mathematically straightforward way to express variability is as an explicit probability density function, and the basic organizing principle for obtaining target detectors is the likelihood ratio test (LRT) [96], which optimally distinguishes between two hypotheses defined by likelihood functions.

For instance, consider the simple case in which a solid (i.e., opaque) target with spectral signature $\mathbf{t}$ covers at least a full pixel. Then, the test of whether a given pixel contains a target is of the form

$$\mathcal{H}_0 : \mathbf{x} = \mathbf{z} \tag{16}$$
$$\mathcal{H}_1 : \mathbf{x} = \mathbf{t}, \tag{17}$$

where $\mathbf{x}$ is the measured spectrum at the pixel of interest, and $\mathbf{z}$ represents the background. If we write $p_t(\mathbf{x})$ as the distribution that corresponds to target variability and $p_z(\mathbf{x})$ as the distribution of the background, then the ratio

$$\mathcal{D}(\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{H}_1)}{p(\mathbf{x}|\mathcal{H}_0)} = \frac{p_t(\mathbf{x})}{p_z(\mathbf{x})} \tag{18}$$

defines an optimal detector for the target in this background. For a given threshold η, $\mathcal{D}(\mathbf{x}) > \eta$ corresponds to the declaration that there is a target at this location (alternative hypothesis $\mathcal{H}_1$), whereas $\mathcal{D}(\mathbf{x}) < \eta$ indicates that the target is absent (null hypothesis $\mathcal{H}_0$). Observing that any monotonic function of a detector is an equivalent detector, we could also use, for instance, $\log \mathcal{D}(\mathbf{x})$. If both $p_t(\mathbf{x})$ and $p_z(\mathbf{x})$ are Gaussian distributions, then the log likelihood detector is, in general, a quadratic function of x [97]. If the means are (nearly) equal but the covariances are different, then a Fukunaga-Koontz [98] detector is called for; the Fukunaga-Koontz transform can also be used for dimension reduction for this quadratic detector [99].

If $p_t(\mathbf{x})$ and $p_z(\mathbf{x})$ are Gaussian distributions with the same covariance, then the optimal detector is linear—indeed, this is the Fisher linear discriminant [97]:

$$\mathcal{D}(\mathbf{x}) = (\mathbf{t} - \mu)^\top R^{-1}(\mathbf{x} - \mu). \tag{19}$$

Here, R is the common covariance, $\mathbf{t}$ is the mean target value, and μ is the mean background value. Although linear detectors are popular and often effective, this argument is a poor motivation, because there is little physical reason to expect the target and background to have the same covariance.

COMPOSITE HYPOTHESIS TESTING

An elegant weapon for a more civilized age.

—Obi-Wan Kenobi (in reference, no doubt, to clairvoyant fusion)

When the distinction has to be made between two families of hypotheses [100], composite hypothesis testing is required. As an example, consider the additive model in (2) as a hypothesis test:

$$\mathcal{H}_0 : \mathbf{x} = \mathbf{z} \tag{20}$$
$$\mathcal{H}_1 : \mathbf{x} = \mathbf{z} + \epsilon t, \text{ with } \epsilon \neq 0. \tag{21}$$

The likelihood ratio leading to an optimal detector is given by

$$\mathcal{D}(\epsilon; \mathbf{x}) = \frac{p_z(\mathbf{x} - \epsilon \mathbf{t})}{p_z(\mathbf{x})}, \tag{22}$$

but, to use this expression, one has to know what value of ϵ to use in the numerator. In general, one does not know ϵ (indeed, if ϵ were actually known, there would be no need to test whether ϵ is nonzero). For this reason, the detector in (22) is called a *clairvoyant detector* [96]. The hypothesis that $\epsilon \neq 0$ is a composite hypothesis because it encompasses a family of simple hypotheses corresponding to specific values of ϵ. This means that the LRT in (22) is insufficient; we do not know ϵ, so we need a detector $\mathcal{D}(\mathbf{x})$ that does not depend on ϵ.

We describe several strategies for producing an ϵ-independent $\mathcal{D}(\mathbf{x})$, but first we consider how to evaluate the quality of such a detector. This is not entirely trivial, either, because the performance of $\mathcal{D}(\mathbf{x})$ does depend on ϵ. For any given ϵ, we know the pdf for the target and background classes and, from that, we can determine the power of a detector (i.e., its detection probability at a given false-alarm rate) and can tell whether one detector is more powerful than another. If the one detector is more powerful than the other for all values of ϵ, then it is said to be *uniformly* more powerful. Furthermore, if there is a detector that is uniformly more powerful than any other detector, then it is a *uniformly-most-powerful* (*UMP*) detector. The UMP detector is the holy grail of composite hypothesis testing; if you can get it, that is the detector you want. Unfortunately, not all problems admit a UMP detector.

One of the few problems that does admit a UMP detector is the additive target model with a Gaussian background distribution. In this special case, the optimal detector $\mathcal{D}(\epsilon;\mathbf{x})$ defined in (22) is effectively independent of ϵ. In particular, for any value of ϵ, one can obtain $\mathcal{D}(\epsilon;\mathbf{x})$ from a simple monotonic rescaling of

$$\mathcal{D}(\mathbf{x}) = \mathbf{t}^\top R^{-1}(\mathbf{x} - \mu), \tag{23}$$

which is the adaptive matched filter (AMF) [101]–[103]. The AMF and the Fisher discriminant in (19) are similar in appearance, differing only in their leading factor—$\mathbf{t}$ for AMF versus $(\mathbf{t} - \mu)$ for Fisher—and the Fisher discriminant is sometimes referred to as a "matched filter" (e.g., in [97]). However, they are derived under quite different assumptions.

A more typical case arises when, for instance, $p_z(\mathbf{x})$ is a multivariate t pdf. One can still derive an expression for $\mathcal{D}(\epsilon; \mathbf{x})$ in that case, but the different values of ϵ will be different detectors; they cannot be rescaled into a common ϵ-independent detector.

The most widely used approach for dealing with the composite hypothesis problem is the generalized LRT (GLRT), in which the variability is parameterized and maximum likelihood estimates of the parameters are sought [96]. In the case of the additive model, the detector becomes $\mathcal{D}(\hat{\epsilon}; \mathbf{x})$, where $\hat{\epsilon}$ is the maximum likelihood estimator:

$$\hat{\epsilon}(\mathbf{x}) = \operatorname{argmax}_\epsilon p_z(\mathbf{x} - \epsilon \mathbf{t}). \tag{24}$$

Thus, the GLRT detector is given by

$$\mathcal{D}(\mathbf{x}) = \mathcal{D}(\hat{\epsilon}(\mathbf{x}); \mathbf{x}) = \frac{\max_{\epsilon} p_z(\mathbf{x} - \epsilon\mathbf{t})}{p_z(\mathbf{x})}. \tag{25}$$

When the background distribution p_z is Gaussian, this likelihood ratio leads trivially to the AMF. If the background distribution is a fatter tailed, elliptically contoured multivariate t-distribution, one can solve (24) explicitly and obtain a closed-form solution for the GLRT detector [104]. In the heavy tailed limit, this GLRT detector becomes the adaptive coherence estimator (ACE) detector:

$$\mathcal{D}(\mathbf{x}) = \frac{\mathbf{t}^{\top} R^{-1}(\mathbf{x} - \mu)}{\sqrt{(\mathbf{x} - \mu)^{\top} R^{-1}(\mathbf{x} - \mu)}}. \tag{26}$$

ACE is a detector that had previously been derived by using different assumptions [105] but that nonetheless has enjoyed remarkable success in many situations that are in clear violation of those initial assumptions [106].

If, instead of the additive target model, we use the replacement target model in (3), then we cannot have a UMP detector. Here, α is the nuisance parameter, and, to obtain a GLRT solution, we must solve

$$\begin{aligned} \hat{\alpha} &= \operatorname{argmax}_{\alpha} p_x(\mathbf{x} \mid \alpha) \\ &= \operatorname{argmax}_{\alpha} (1-\alpha)^{-d} p_z\left(\frac{\mathbf{x} - \alpha\mathbf{t}}{1-\alpha}\right). \end{aligned} \tag{27}$$

It turns out that this can be solved in closed form when p_z is Gaussian [107], leading to the finite target matched filter (FTMF) detector. The result has been further extended to a family of elliptically contoured background distributions [108], [109].

A generalization of the GLRT concept is the *clairvoyant fusion* (*CF*) approach [110], [111], in which one begins with a clairvoyant detector—that is, the detector $\mathcal{D}(\epsilon; \mathbf{x})$ that is optimal if ϵ is the nonzero value in the alternative hypothesis. Then, we can express the result in (25) by writing

$$\mathcal{D}(\mathbf{x}) = \max_{\epsilon} \mathcal{D}(\epsilon; \mathbf{x}). \tag{28}$$

We observe that this GLRT detector is a "max-fusion" over clairvoyant detectors. However, if $\lambda(\epsilon)$ is any positive function of ϵ, then $\mathcal{D}^*(\epsilon; \mathbf{x}) = \lambda(\epsilon)\mathcal{D}(\epsilon; \mathbf{x})$ is also a clairvoyant detector, because it is just a monotonic rescaling. We can now perform a max-fusion over this new family of clairvoyant detectors to create a new CF detector:

$$\mathcal{D}(\mathbf{x}) = \max_{\epsilon} \mathcal{D}^*(\epsilon; \mathbf{x}) \tag{29}$$

$$= \max_{\epsilon} \lambda(\epsilon) \mathcal{D}(\epsilon; \mathbf{x}) \tag{30}$$

$$= \frac{\max_{\epsilon} \lambda(\epsilon) p_z(\mathbf{x} - \epsilon\mathbf{t})}{p_z(\mathbf{x})}. \tag{31}$$

Because the choice of $\lambda(\epsilon)$ is virtually limitless (it only matters that it is positive for all values of ϵ), the CF framework provides great flexibility in deriving a detector. There is, unfortunately, little guidance with respect to the "best" choice of the function $\lambda(\epsilon)$, but a clever practitioner may be able to craft a $\lambda(\epsilon)$ that enables the inversion

$$\hat{\epsilon}(\mathbf{x}) = \operatorname{argmax}_{\epsilon} \lambda(\epsilon) \mathcal{D}(\epsilon; \mathbf{x}) \tag{32}$$

to be analytically tractable, thus leading to a closed-form solution for the detector.

When prior probability density functions are available, Bayesian composite hypothesis testing can be performed [100], [112]. Here, instead of taking a maximum over clairvoyant detectors, one takes a weighted average. Again, using the additive target model as an example, we have, in contrast to (25) or (31),

$$\mathcal{D}(\mathbf{x}) = \frac{\int_{\epsilon} d\epsilon\, \pi(\epsilon) p_z(\mathbf{x} - \epsilon\mathbf{t})}{p_z(\mathbf{x})}, \tag{33}$$

where $\pi(\epsilon)$ is the prior; it is an arbitrary nonnegative function of ϵ chosen by the practitioner.

None of these three approaches—GLRT, CF, or Bayesian—can be counted on to produce a UMP detector, because such detectors very often simply do not exist. However, when UMP is not available, a next-most-desirable trait in a detector is that it be admissible. A detector is admissible if there does not exist another detector that is uniformly more powerful than it is. Although the integral in (33) often makes Bayesian detectors less convenient than GLRT or CF detectors associated with closed-form expressions, Bayesian detectors have the following important theoretical advantage: they are provably admissible [100]. By contrast, some GLRT and CF detectors are not admissible [111]. The notion of "admissible" detectors is of great theoretical importance, but it is often remarked that perfectly respectable (i.e., useful) detectors may still be formally inadmissible.

Although beyond the scope of this review, a natural follow-up would be to determine the sensitivity of different algorithms to material variability; algorithms that are more "sophisticated" (i.e., more finely tuned to optimizing performance with respect to a special case) may be less robust to deviations from those special cases. In practice, the simpler algorithms may perform better.

INVARIANT APPROACHES

By imposing the constraint on a target detector that it be explicitly invariant to some aspect of that target's variability, we obtain an effective robustness to that variability. This means that we are allowed to have a given amount of ignorance of the variables that alter the target spectrum amplitude and shape [113]. Formally, if we consider that the target spectrum $\mathbf{t}$ depends on some variable parameters enclosed in the vector $\mathbf{\Theta}$, an invariant detector should ideally produce the same output (in terms of declaring $\mathcal{H}_0$ or $\mathcal{H}_1$) regardless of $\mathbf{\Theta}$. This condition is ensured only in some special cases (which will be discussed) [114]. In many other cases, the condition is relaxed to obtain a detector that is robust (though not strictly invariant) to target variability. In striving toward obtaining invariant detectors, researchers have invoked a variety of models, as described in the "Models for Spectral Variability" section.

Through the exploitation of probabilistic models (see the "Probabilistic Models" section), GLRT invariance has been widely investigated [100], [113]–[116]. The GLRT has been found to be invariant (in terms of the condition stated earlier) with respect to a group of transformations if, upon transformation, the pdfs conditioned to the two hypotheses remain in the same family and the parameter spaces are preserved [100], [114]. Formally, we can write [114]

$$\begin{cases} f \in \mathcal{F} \\ p(\mathbf{x}; \boldsymbol{\Theta}) = p(f\mathbf{x}; f\boldsymbol{\Theta}) \\ f: \Omega \to \Omega \end{cases} \Rightarrow \mathcal{D}(f\mathbf{x}) = \mathcal{D}(\mathbf{x}), \tag{34}$$

where $\boldsymbol{\Theta}$ is the parameter vector that belongs to the parameter spaces Ω_0 and Ω_1 under the null and alternative hypotheses, respectively, and $f \in \mathcal{F}$ is the set of transformations with respect to which the GLRT is invariant. A proof is given in [114]. In [115], the relationships between the GLRT and the UMP invariant (UMPI) tests are described. Other approaches to find UMPI tests are illustrated in [113].

Aside from the GLRT and UMPI tests, the most widely used invariant approaches are invariant in a broader sense, in that they are robust to variability (i.e., their performances are not as degraded as those of conventional detectors) while not strictly ensuring the same detector output upon variation of the target signature. Nonetheless, in the literature they are called *invariant methods*, and we will adhere to this nomenclature.

One straightforward way to impose invariance is by invoking (unconstrained) subspace models (see the "Unconstrained Subspace Models" section). If $\mathcal{T}$ is a subspace, then it can be parameterized by a matrix $\mathbf{M}$, so that $\mathbf{t} \in \mathcal{T}$ is equivalent to the existence of a vector of coefficients $\mathbf{p}$ with $\mathbf{t} = \mathbf{M}\mathbf{p}$.

One general approach for finding this target subspace is to obtain (by measurement or simulation or both) a discrete (and ideally large) set $\mathcal{T} = \{\mathbf{t}^{(i)}\}_{i=1}^{C}$ of radiance spectra for a given target material. This set should span the range of variation over which robustness is desired, and it can include intrinsic, extrinsic, and/or environmental variability [13], [35], [36], [45]–[47], [117]. For example, Figure 4 illustrates spectral variability under a range of atmospheric conditions by producing such an ensemble of individual spectra.

The idea of using a low-dimensional subspace is that, for some $Q \ll C$, we can find a model characterized by a $Q \times d$ matrix $\mathbf{M}$ which spans a subspace that approximates each of the target spectra in the set, that is,

$$\mathbf{t}^{(i)} \approx \sum_{k=1}^{Q} p_k^{(i)} \mathbf{m}_k = \mathbf{M}\mathbf{p}^{(i)}. \tag{35}$$

From the subspace defined by $\mathbf{M}$, we can invoke, for instance, subspace versions of the AMF or ACE [97], [116]. This approach has been extensively pursued in the literature [13], [35], [36], [45]–[47], [71], [117], [118]. Here, the target radiance spectra have generally been synthetically generated by radiative transfer modeling, and variability has been incorporated by varying the environmental, acquisition geometry, and illumination parameters, together with variations of the BRDF, by exploiting suitable physics-based models [13] and, in some cases, also exploiting in-scene information [117] or large spectral library databases [3].

The need to identify a low-dimensional linear subspace poses the perennial challenge of estimating the "optimal" subspace dimensionality. If the dimension is too small, then the approximation in (35) becomes poor, but as the dimension increases, the subspaces can harbor ever larger numbers of false alarms [71]. Another general drawback of this approach is that, regardless of its dimensionality, the size of the subspace spanned by the basis vectors is not constrained because the coefficients $p_k^{(i)}$ can take any (even unphysical) values [119].

One way to deal with subspace high-dimensionality issues is to constrain the subspace by imposing conditions on the coefficients $p_k^{(i)}$, as, for instance, is done with constrained subspace or simplex modeling (see the "Constrained Subspace Models" section). Various approaches have been explored for simplex-based variable target detection [71], [119], [120], e.g., by enforcing only the additivity constraint [71], enforcing both constraints [119], or performing simultaneous linear unmixing on the target and background spectra together to obtain physical abundances [120]. This has led to methods such as simplex ACE or simplex AMF [71], which have exhibited detection performances that, in contrast to their unconstrained subspace-based counterparts, are stable with respect to increasing amounts of spectral variability and increasing sizes of the target subspace.

More recent approaches to building invariant algorithms resort to nonlinear models, such as manifolds (see the "Topological Manifolds" section). As previously noted, not all variability effects on spectra can be adequately captured using low-dimensional linear models, but the variable target spectra often "live" in a lower-dimensional nonlinear manifold. In [89], the target radiance spectra $\{t^{(i)}\}_{i=1}^{C}$ are assumed to lie within a manifold, and a graph-based manifold learning approach is used to perform invariant target detection. Resorting to nonlinear manifolds is often necessary in underwater hyperspectral remote sensing as well, where water spectra change nonlinearly upon variations of depth and water-inherent optical properties [81]. Nonlinear dimensionality reduction by manifold learning has been successfully exploited in several hyperspectral underwater remote sensing applications, such as underwater object detection [81] and hyperspectral bathymetry [90].

One challenge with using radiative transfer modeling to create a very large set of variable target spectra is that the computation can be prohibitive. It is worth mentioning that some "emulation" approaches have been proposed, in which a reduced number of available variable target spectra is used to learn a stochastic model function of a given number of predictor variables. The learned model is then used to generate an arbitrary number of variable target spectra with potential real-time applications [121].

MATCHED-PAIR MACHINE-LEARNING APPROACH

As is evident in, for instance, the additive and replacement models, the observed spectrum at a given pixel is often a combination of target and background. As a consequence, target variability is often mixed up with background variability. Furthermore, although target variability can be informed by *ab initio* physics modeling, the background variability is best estimated by looking at actual background pixels. The aim of the matched-pair machine-learning (MPML) framework [109], [122] is to produce a target detection algorithm customized to the observed background while at the same time exploiting a model of target variability and how the target and background interact. In MPML, every background pixel $\mathbf{z}$ is paired with a background-plus-target pixel $\mathbf{x} = \mathbf{z} + \epsilon \mathbf{t}$ (in the case of the additive target model), and a training set of these matched pairs is used to construct a machine-learning classifier to distinguish background from background-plus-target. Spectrally variable targets can be dealt with very naturally in this framework; the $\mathbf{t}$ that is used to create the target-present half of the matched pair is drawn from a distribution $p_t(\mathbf{t})$ that characterizes target spectral variability.

In fact, MPML can be run in a "transductive" mode [123], in which the classifier is trained on the same data to which it is applied. Normally this is problematic, but it works here because MPML does not actually use labeled data. When the classifier is applied to the original hyperspectral image pixels, it will identify those pixels that are most like the artificial background-plus-target pixels in the training set; that is, it will find pixels in the original image that are likely to have target in them. There is an implicit assumption that the vast majority of the pixels in the original image are target-free background pixels, so the contamination effect of the few target-containing pixels in the training step will be minimal. It is important that the classifier is not overfit to the data (and cross-validation is useful for assessing that); an overfit classifier will classify all the original pixels as target-free, even those few that do have target.

OTHER MACHINE-LEARNING APPROACHES

It is expected that machine-learning approaches, and nonparametric fitting in general, will continue to garner attention as tools for modeling variability. Hyperspectral target detection is enhanced by using these tools to characterize the background in a local [124] or context-dependent [125], [126] manner.

ROBUSTNESS TO VARIABILITY IN OTHER HYPERSPECTRAL IMAGE ANALYSIS TASKS

Although we have focused here on robustness to variability for target detection, many efforts can be found throughout the hyperspectral image analysis literature for coping with the spectral variability of materials. Endmember extraction and unmixing methods, for instance, in their simplest forms, assume "pure" endmembers; however, for good performance on more realistic hyperspectral data, algorithms require robustness to variability within the endmember classes [74], [127], [128]. Clustering and material discrimination methods have also been developed to handle intraclass spectral variability [128], [129]. The variability due to noise and bottom-reflectance materials has been taken into account in retrievals of subsurface reflectance in shallow-water remote sensing [130]. Change-detection algorithms require suppressing the pervasive differences between images (due to environmental effects, sensor calibrations, shadows, and so on) to identify the interesting changes [131]–[133].

MITIGATING SPECTRAL VARIABILITY IN HYPERSPECTRAL DATA

Instead of designing exploitation methods that are robust to variability, sometimes it may be worth trying to remove or mitigate the variability prior to proceeding with the hyperspectral image exploitation task. In the literature, this approach has mostly been pursued to back out environmentally induced variability, such as atmospheric and illumination effects [134], shadows [32], [135], and clouds [136], but approaches aimed at mitigating variability due to material morphology and composition have also been proposed [137], [138]. In the following sections, we provide a brief overview of these approaches.

ATMOSPHERIC COMPENSATION

The most widely used preprocessing step aimed at removing environmentally induced variability is undoubtedly AC. Literally, *AC* refers to the procedure of obtaining an approximate ground-leaving reflectance image starting from the calibrated spectral radiance hyperspectral data cube [134]. This approach is generally preferred by image analysts, such as geologists or spectroscopists, although in some applications for which a target material needs to be searched for within a hyperspectral image, the so-called forward modeling AC approach of converting the spectral library reflectance into an at-sensor radiance spectrum may be pursued [36]. AC methods can be broadly divided into two categories: empirical scene-based methods [139]–[144] and radiative transfer modeling-based methods [38], [134], [145]–[147], although some hybrid approaches may be found as well [148]. Empirical methods exploit data directly from the radiance image, such as the average spectrum of the scene [139], the spectrum of a "spectrally flat" material in the scene [143], or vegetation and shade spectra [149]. Among these, the popular empirical line method (ELM) [142], [144] requires field measurements of reflectance spectra for at least one dark and one bright reference material and linearly regresses the imagery radiance data against the reflectance field measurements; this enables the derivation of gain and offset curves, which are then applied to each image pixel. Like most empirical methods, ELM does not require absolute radiometric calibration, but it does require further adjustments if the atmospheric and illumination conditions are not stationary within the scene [150].

Whereas empirical methods are widely applied in operational scenarios [151], over the years researchers have put their efforts toward "forever improving the physical model" [152] to provide increasingly more accurate radiative-transfer-modeling-based AC techniques. The AC approaches belonging to this category are based on an analytic model for the radiation transfer in the atmosphere, such as that explored in the "Environmentally Induced Target Variability" section, and, thus, need to estimate the various model parameters and then invert the model to retrieve the surface reflectance. Model parameter estimation is generally accomplished by resorting to radiative-transfer codes, such as MODTRAN [54] or 5S [153], or exploiting the imaged radiance in certain bands to estimate visibility, spatially variable parameters (e.g., WV and aerosols), and adjacency effects. One of the first AC techniques led to the atmospheric removal program (ATREM) [145], which did not explicitly account for adjacency effects. This was instead incorporated in the approach of fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH) [154], which is a very popular commercial AC software package. More recently, the atmospheric and topographic correction algorithm (ATCOR) [147] has been developed, which also features a correction for topographic effects (provided that a digital elevation map of the scene is available).

SHADOW AND CLOUD REMOVAL

Shadows and clouds are among the most nonstationary sources of variability in hyperspectral imagery, in both space and time. Whereas the effects of clouds are mostly significant for high-altitude acquisitions (e.g., from spaceborne sensors), the impacts of shadows are stronger for high spatial resolution imagery, especially in the urban environment.

Approaches to shadow removal are typically based on specific transformations that are applied to the imaged data [155], [32]. For instance, a transformation from Cartesian to hyperspherical coordinates allows the data to be separated into $d - 1$ angular components related to spectral information content and one radial component whose magnitude represents the overall brightness level [32]. By segmenting the image and identifying those clusters with the lowest magnitude of the radial component, a shadow mask can be obtained [32]. Another approach to remove shadows is to exploit ancillary data, such as lidar 3D point clouds [44], [135].

Detection of clouds is essentially based on finding pixels in which the cloud optical depth is greater than a specific threshold [136] or evaluating a combination of threshold tests [156]. This enables generation of a cloud mask. When possible, the radiative effects (e.g., changes in illumination, shadows, temperature variations, spatial structure, and enhanced adjacency effects) of cloud layers are corrected [136].

SPECTRAL REFLECTANCE DECOMPOSITION WITH MULTIPLE GAUSSIAN CURVES

A physics-inspired spectral characterization for material reflectance spectra was presented by Lanker et al. [137], with the aim of suppressing variability due primarily to material morphology and composition. Inspiration was drawn from the Lorentz oscillator model applied to the material dielectric function (i.e., how, as a function of wavelength, the optical index of refraction varies). Each Lorenz oscillator, roughly speaking, corresponds to a spectral "line" (although for solid materials, unlike gases, individual lines can be very broad); thus, a model of material spectral reflectance with multiple Lorentz oscillators identifies multiple lines in a spectrum. In [137], the authors chose to fit the curve of reflection versus wavelength using Gaussian line shapes—this was found to be simpler and less computationally intense than fitting the complex dielectric function with Lorentz oscillators. This is similar to the curve-fitting approaches performed in mineral spectroscopy [157].

It was experimentally found that, although the Gaussian amplitudes were subject to considerable variations upon varying atmospheric, illumination, and viewing conditions as well as material morphology and composition, the locations and widths of the Gaussian functions were much less affected [137], [138]. These findings suggest that alternative profiles based on the locations and widths of the Gaussian functions used to fit the spectra could be adopted for robust characterization of target material spectra and potentially for background spectral variability as well.

IN-SCENE TARGET SIGNATURE CHARACTERIZATION

In some circumstances, the problem of target variability can be bypassed by directly estimating from the image the most discriminative target spectral signature for a given scenario and material by means of a multiple-instance machine-learning approach [158]. This can be accomplished when at least imprecisely labeled training samples are available—for instance, when GPS coordinates for the targets are known and general regions of pixels containing the targets can be identified. (GPS precision is generally limited by the coregistration accuracy; thus, pixel-level ground truthing is hardly ever achieved.) Using a simple iterative algorithm with a closed-form solution for the update rule, the multiple-instance learning approach is capable of retrieving the target signature that maximizes the cosine spectral similarity (more robust in the case of mixed training data) between the estimated signature and the positive (target-including) instances while minimizing the similarity with the non-target-labeled instances. As a byproduct, the estimated signature helps uncover the discriminative spectral characteristics and features of the target class, which are otherwise deeply buried within the mixed and imprecisely labeled training pixels [158].

CONCLUSIONS

Only a Sith deals in absolutes.

—Obi-Wan Kenobi

Spectral variability fundamentally underlies the whole problem of analyzing, understanding, and interpreting hyperspectral imagery. Of course, spectral variability is not all bad. It is the variability between different materials that allows us to distinguish them in spectral remote sensing. In this discussion, we have emphasized designing algorithms that are robust to variability, but, for some applications, we might want algorithms that are sensitive to variability, even within the same material. For instance, in our search for the perfect beach, we may want to find sand of a very particular type, not just sand as a single class.

Our aim in analyzing hyperspectral imagery is to enhance the distinctions we care about (natural versus manmade, healthy vegetation versus senescent, target versus background, etc.) while suppressing the distinctions that needlessly confound our analysis (sunlit versus shadow, humid atmosphere versus dry, nadir versus slant angle view, and so on), with the realization that *what matters* and *what distracts* both depend on the application at hand.

In this survey, we have discussed many of the causes of spectral variability that complicate algorithms for hyperspectral analysis, described some of the models used to characterize that variability, and reviewed a number of strategies for dealing with this variability. In many cases, those strategies are a direct consequence of having an explicit model for the variability.

The spectral variability of materials is a topic of relevance for many aspects of hyperspectral image analysis. It arises in landcover classification, crop health characterization, image segmentation, endmember extraction, spectral unmixing, change analysis, and target detection. We argue that many of the approaches for dealing with spectral variability in target detection carry over to other problems, although their use in target detection is often more straightforward and illustrative.

Sorry about the mess.

—Han Solo

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